MODULE 04 / LESSON 01

# Exploratory Data Analysis

Describe a dataset, expose quality problems, and form questions before modelling.

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WHY THIS MATTERS

# Start from what you already know

## DataFrames, filtering, grouping, and missing values

01 Explain data types in plain language.

02 Use descriptive statistics in a small worked example.

03 Complete the guided lab and verify correlation and causation.

CORE CONCEPT

# The vocabulary of this topic

Describe a dataset, expose quality problems, and form questions before modelling.

**data types: Data types define the values an operation can accept and the behavior that operation will produce.**

**descriptive statistics: Descriptive statistics summarize the center, spread, frequency, and extremes of observed values.**

**distributions: A distribution reveals center, spread, skew, modes, and extremes that a single average cannot show.**

**correlation and causation: Correlation measures association; causation requires evidence that changing one factor changes another.**

FOUR CONNECTED IDEAS

# How the concept works

01

**data types**

→

02

**descriptive statistics**

→

03

**distributions**

→

04

**correlation and causation**

EDA moves from schema and row grain to quality, center, spread, shape, relationships, and documented findings.

CORE CONCEPT

# Read every part of the mechanism

EDA moves from schema and row grain to quality, center, spread, shape, relationships, and documented findings.

**mean: Sum divided by count.**

**median: Middle ordered value.**

**distribution: How frequently values occur.**

WORKED EXAMPLE

# Worked example by hand

**For incomes [20, 22, 24, 26, 200], compare mean 58.4 with median 24 and explain the outlier.**

01

### Known values

02

### Apply the mechanism

03

### Interpret the result

RUN IT AND INSPECT THE RESULT

# Map the concept to Python

```
values = pd.Series([20, 22, 24, 26, 200])
print(values.mean(), values.median())
```

OUTPUT

```
58.4 24.0
```

**Each line corresponds to a concept already explained.**

QUALITY GATE

# Build the complete mental model

- ✓ Use median and interquartile range when extreme values distort mean and standard deviation.
- ✓ A histogram shows distribution shape; a box plot emphasizes quartiles and potential outliers.
- ✓ Compare target rates across meaningful groups before proposing a predictive feature.
- ✓ EDA produces testable data-quality and modelling hypotheses, not final causal conclusions.

FOUR CONNECTED IDEAS

# Notebook readiness map

01 **data types** →

02 **descriptive statistics** →

03 **distributions** →

04 **correlation and causation**

Use the concepts from this lesson to read and complete https://github.com/curiousily/AI-Bootcamp/blob/master/02.exploratory-data-analysis.ipynb.

RISK REVIEW

# Common beginner mistakes

01 Treating `unknown` as a valid category without investigation.

02 Claiming that correlation proves one variable causes another.

GUIDED LAB

# Profile the Bank Marketing data and support three findings with statistics and plots.

01

**Predict**

02

**Run**

03

**Inspect**

04

**Explain**

SESSION SYNTHESIS

# Keep the system, not isolated definitions.

01 **Explain data types in plain language.**

02 **Use descriptive statistics in a small worked example.**

03 **Complete the guided lab and verify correlation and causation.**

NEXT EVIDENCE

**An EDA report covering schema, missing/sentinel values, distributions, class balance, relationships, and limitations.**

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